

# APPLICATIONS OF COMBINATORIAL AND TOPOLOGICAL GAMES IN CLASSICAL MATHEMATICS

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For the UT San Antonio Mathematics student research seminar

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## INTRODUCTION

In 1935 Stanisław Mazur wrote a problem in the Scottish Book, the notebook kept at the Scottish Café in Lwów where the Polish mathematicians of the time did a good part of their thinking, which is proof that good mathematics and fine coffee have always gone together. Two players take turns choosing nested intervals of the real line, each inside the previous one, and the question is when the first player can force the point where the intervals close down to land in a given set. Banach found the answer, and Oxtoby published it in 1957. The game has been called the Banach–Mazur game ever since.

It looks like a party game, and in a sense it is one, although admittedly a strange party. But it turns out to be a remarkably good way of thinking about *how big* a set is. A set is “large” if one of the players can always steer the game into it, and “small” if the other player can always steer the game away from it. Over the last sixty years people have changed the rules in all sorts of ways (prescribing the sizes of the moves, limiting how much the players may compute, letting them play vectors, points or covers instead of intervals), and each new

version has cracked open a problem in some unexpected corner of mathematics. These notes tell eleven of those stories.

Part I is about the Banach–Mazur game itself and its close relatives, the games of Choquet type, in which the two players take turns shrinking a set:

**Number theory:** Schmidt’s games and badly approximable numbers (Schmidt, 1966).

**Dynamical systems and Lie theory:** Bounded orbits of flows on spaces of lattices: Margulis’s conjecture, proved by Kleinbock and Margulis (1996).

**Theoretical computer science:** Baire category inside complexity classes (Mehlhorn, 1973; Lutz, 1990).

**Topological groups:** When separate continuity implies joint continuity (Ellis, 1957; Namioka, 1974; Kenderov, Korteov and Moors, 2001).

**Functional analysis:** Asplund spaces (Namioka–Phelps and Stegall, 1975; the game of Kenderov and Moors, 1999) and Gowers’s dichotomy for Banach spaces (Gowers, 1996).

**Infinite-dimensional Ramsey theory:** The Ellentuck topology (Ellentuck, 1974) and the Kastanas game (Kastanas, 1983).

Part II is about other famous topological games, mentioned by Frank, in which the players do not simply shrink a set:

**Set-theoretic topology:** The Menger game and the refutation of Menger’s conjecture (Fremlin and Miller, 1988).

**Descriptive set theory:** Borel determinacy (Martin, 1975) and the Wadge hierarchy (Wadge, 1972).

The Appendix collects three beautiful applications of games that are not topological at all:

**Probabilistic combinatorics:** The Erdős–Selfridge theorem and derandomization (Erdős and Selfridge, 1973).

**Fixed point theory:** Hex and the Brouwer fixed point theorem (Gale, 1979).

**Model theory:** Ehrenfeucht–Fraïssé games (Fraïssé, 1954; Ehrenfeucht, 1961).

A word of advice before we start. Part I climbs some steep mountains (Lie groups, Banach spaces, infinite-dimensional Ramsey theory), and nobody expects you to enjoy every view on the first attempt. If you ever feel dizzy, jump to the Appendix! Hex, Maker–Breaker games and Ehrenfeucht–Fraïssé games are played on finite boards or with finitely many moves, they need no topology at all, and they give you the same intuition about strategies with none of the baggage. Then come back.

Here, for the record, is the game that started it all, in the form in which we will keep modifying it. Every game in Part I is this game with a few rules changed.

**The Banach–Mazur game.** Fix a set  $S \subseteq \mathbb{R}$ . Player I chooses a closed interval  $U_0$  of positive length; Player II chooses a closed interval  $V_0 \subseteq U_0$  of positive length; Player I chooses  $U_1 \subseteq V_0$ ; and so on, forever. Player II wins if every point of  $\bigcap_k V_k$  belongs to  $S$ . Theorem (Banach; Oxtoby, 1957): Player II has a winning strategy if and only if  $\mathbb{R} \setminus S$  is meager, that is, if and only if  $S$  is “large” in the sense of Baire category.

Two features of games come back again and again. First, a winning strategy is built one move at a time, so a big global question (“is this set large?”) turns into a small local one (“what should I play next?”). Second, strategies can be *combined*: a player can pursue several goals at once by taking turns between them, one round for this goal, the next round

for that one. This is why sets defined by games survive countable intersections, something that measure and dimension simply do not do.

*Conventions.* We follow our notes on the Banach–Mazur and Choquet games. Player I moves first, Player II answers inside Player I’s move, and the payoff set belongs to Player II: she is the one trying to produce an object with a given property, and he is the one trying to stop her. Player I is the villain of every story in these notes, which is unfair to him, since he only wants to see whether Player II really knows what she is doing. Where an author used other names for the players (Spoiler and Duplicator, Maker and Breaker, ONE and TWO,  $\beta$  and  $\alpha$ ), we say which of them is our Player I. Each section has the same shape: the problem (in a pink box), its history, the main result (in a blue box), the game, and how the game solves the problem.

## Part I. Sculptors: Banach–Mazur and Choquet games

### 1. APPLICATIONS TO DIOPHANTINE APPROXIMATION

**The problem.** Every real number can be approximated by fractions, but some numbers are much harder to approximate than others. For  $t \in \mathbb{R}$  let  $\|t\|$  be the distance from  $t$  to the nearest integer. A vector  $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$  is *badly approximable* if there is a constant  $c = c(\mathbf{x}) > 0$  such that

$$\max_{1 \leq i \leq n} \|qx_i\| \geq cq^{-1/n} \quad \text{for every integer } q \geq 1.$$

For  $n = 1$  this says  $|x - p/q| \geq c/q^2$  for every fraction  $p/q$ . Let **Bad**( $n$ ) be the set of badly approximable vectors. Dirichlet’s theorem (1842) says that  $\max_i \|qx_i\| < q^{-1/n}$  for infinitely many  $q$ , so badly approximable vectors are the ones for which Dirichlet’s theorem cannot be improved by more than a constant: they are the most stubborn vectors there are. How big is **Bad**( $n$ )? In particular, is it still big after we intersect countably many translates of it?

✂ 1.1. **History of the problem and its solution using topological games.** For  $n = 1$  there is a pretty description: a real number is badly approximable exactly when the partial quotients of its continued fraction are bounded. So the golden ratio  $(1 + \sqrt{5})/2 = [1; 1, 1, 1, \dots]$ , whose partial quotients are as small as they can be, is the most badly approximable number of all. But are badly approximable numbers common or rare? The answer is “both”. Khintchine proved in 1924 that **Bad**( $n$ ) has Lebesgue measure zero, so a number chosen at random is almost surely *not* badly approximable. On the other hand, Jarník proved in 1928 that **Bad**(1) has Hausdorff dimension 1, the largest possible. So **Bad**(1) is tiny for measure and huge for dimension.

Neither measure nor dimension says anything useful about intersections: two sets of full dimension can be disjoint. So a question as innocent as “is there a number  $x$  such that  $x + \sqrt{2}$ ,  $x + \sqrt{3}$ ,  $x + \sqrt{5}$ ,  $\dots$  are all badly approximable?” was out of reach.

The problem was solved by Wolfgang Schmidt in 1966 [Sch66]. He invented a new version of the Banach–Mazur game, now called Schmidt’s game, proved that its winning sets have full Hausdorff dimension and survive countable intersections, and showed that **Bad**(1) is a winning set. In 1969 he did the same for **Bad**( $n$ ) for every  $n$  [Sch69]. The game kept paying dividends. In 1983 Schmidt conjectured that for weights  $i, j > 0$  with  $i + j = 1$ , the sets **Bad**( $i, j$ ) and **Bad**( $j, i$ ) of vectors in  $\mathbb{R}^2$  that are badly approximable with weights  $(i, j)$  and  $(j, i)$  have a point in common, and he observed that a counterexample would

prove the famous Littlewood conjecture. Badziahin, Pollington and Velani proved Schmidt's conjecture in 2011 [BPV11], and in 2016 An showed that each  $\mathbf{Bad}(i, j)$  is winning for Schmidt's game [An16], which gives the conjecture, in a stronger form, from the intersection property.

*Recognition.* Schmidt received the Cole Prize in Number Theory in 1972, and he is one of the few number theorists invited to address the International Congress of Mathematicians three times. The solution of Schmidt's conjecture by Badziahin, Pollington and Velani appeared in the *Annals of Mathematics*.

**1.2. The Main Result of this section.** The main result consists of Schmidt's theorems on his game and on badly approximable vectors. The word " $\alpha$ -winning" is explained in the next subsection.

**Theorem 1.1 (Schmidt, 1966 and 1969).** *Let  $0 < \alpha < 1$ .*

- (1) *If  $S \subseteq \mathbb{R}^n$  is  $\alpha$ -winning, then  $S \cap U$  has Hausdorff dimension  $n$  for every non-empty open set  $U \subseteq \mathbb{R}^n$ .*
- (2) *The intersection of countably many  $\alpha$ -winning subsets of  $\mathbb{R}^n$  is  $\alpha$ -winning.*
- (3)  *$\mathbf{Bad}(1)$  is  $\frac{1}{2}$ -winning, and  $\mathbf{Bad}(n)$  is  $\alpha$ -winning for suitable  $\alpha > 0$ .*

A translate of an  $\alpha$ -winning set is  $\alpha$ -winning (just translate the strategy), so by (2) the set  $\bigcap_k (\mathbf{Bad}(n) + \mathbf{t}_k)$  is  $\alpha$ -winning for every sequence of vectors  $\mathbf{t}_k$ , and by (1) it has full dimension. In particular, there are (lots of) numbers  $x$  such that  $x + \sqrt{p}$  is badly approximable for every prime  $p$ .

**1.3. The game.** Let  $(X, d)$  be a complete metric space and fix two numbers  $\alpha, \beta \in (0, 1)$ . Schmidt called the players White and Black; in Choquet's terminology our Player I is  $\beta$  and our Player II is  $\alpha$ . The game is played with closed balls, and the radii are fixed in advance.

- (1) Player I chooses a closed ball  $U_0 = \overline{B}(x_0, r_0)$  of any radius he likes.
- (2) Player II chooses a closed ball  $V_0 \subseteq U_0$  of radius  $\alpha r_0$ .
- (3) Player I chooses a closed ball  $U_1 \subseteq V_0$  of radius  $\beta \alpha r_0$ .
- (4) And so on: the play produces nested balls  $U_0 \supseteq V_0 \supseteq U_1 \supseteq V_1 \supseteq \dots$ , where  $U_k$  has radius  $r_k = (\alpha\beta)^k r_0$  and  $V_k$  has radius  $\alpha r_k$ .

The radii go to 0 and  $X$  is complete, so the balls close down on a single point  $x_\infty$ . Player II wins if  $x_\infty \in S$ . The set  $S$  is  $\alpha$ -winning if Player II has a winning strategy for every  $\beta \in (0, 1)$ . Notice the order of the quantifiers: Player II declares her handicap  $\alpha$  first, promising to shrink by the factor  $\alpha$  every time, and only then does Player I pick  $\beta$ , as aggressive as he likes. Her strategy is allowed to depend on  $\beta$ , but  $\alpha$  is fixed once and for all. This is what makes the notion robust enough to survive countable intersections. On the real line the balls are just closed intervals, and it is convenient to talk about their lengths instead of their radii.

Compared with the Banach–Mazur game, the only new rule is that the sizes of the moves are prescribed. That small change is what makes the game see Hausdorff dimension: Player I can no longer shrink everything to a point in one move.

**1.4. How the game solves the problem.** We go through the three parts of Theorem 1.1, starting with the one that is pure number theory. For simplicity we prove that  $\mathbf{Bad}(1)$  is  $\alpha$ -winning for every  $\alpha < \frac{1}{2}$ ; Schmidt's sharper bookkeeping also handles  $\alpha = \frac{1}{2}$ .

**Badly approximable numbers: the plan.** Fix  $\alpha < \frac{1}{2}$  and  $\beta \in (0, 1)$ , and let Player I open with an interval  $U_0$  of length  $\rho_0$ . Put  $R = 1/(\alpha\beta) > 1$ , so that every round shrinks the intervals by the factor  $R$ : the interval  $U_k$  has length  $\rho_k = \rho_0 R^{-k}$ . We will find a constant  $c > 0$  and a strategy for Player II such that the final point  $x$  satisfies

$$\left| x - \frac{p}{q} \right| \geq \frac{c}{q^2} \quad \text{for every fraction } p/q \text{ with } q \geq 1.$$

Call the interval  $(\frac{p}{q} - \frac{c}{q^2}, \frac{p}{q} + \frac{c}{q^2})$  the *no-go zone* of  $p/q$ . Player II's job is to keep the final point out of every no-go zone. There are infinitely many zones, but she has infinitely many moves, and the whole proof is about giving each move a manageable amount of work.

**Idea 1: fractions with small denominators keep their distance.** If  $p/q \neq p'/q'$ , then

$$\left| \frac{p}{q} - \frac{p'}{q'} \right| = \frac{|pq' - p'q|}{qq'} \geq \frac{1}{qq'},$$

because  $pq' - p'q$  is a non-zero integer. So two fractions whose denominators are both small cannot be close together. This little inequality is the engine of the whole argument.

**Idea 2: sort the fractions into generations.** Let  $A = 1/(3\rho_0)$ . Say that  $p/q$  belongs to *generation 0* if  $q^2 < A$ , and to *generation  $k \geq 1$*  if

$$AR^{k-1} \leq q^2 < AR^k.$$

Every fraction belongs to exactly one generation. Think of the generations as the octaves of a piano: we group the denominators by size, one octave per round, so that within a single octave all the fractions keep a respectful distance from one another. Generation  $k$  is the one Player II will deal with in round  $k$ . By Idea 1, two different fractions of generation  $k$  are at distance more than  $1/(AR^k) = 3\rho_0 R^{-k} = 3\rho_k$ . Now fatten Player I's interval  $U_k$  by  $\rho_k$  on each side; the fattened interval has length  $3\rho_k$ , so it contains *at most one* fraction of generation  $k$ . If there is one, call it the *dangerous fraction* of round  $k$ .

**Idea 3: the no-go zones are small.** A fraction of generation  $k \geq 1$  has  $q^2 \geq AR^{k-1}$ , so its no-go zone has length

$$\frac{2c}{q^2} \leq \frac{2c}{AR^{k-1}} = 6cR\rho_k.$$

Choose  $c$  so small that  $6cR \leq 1 - 2\alpha$  and  $2c \leq (1 - 2\alpha)\rho_0$  (the second condition takes care of generation 0, whose zones have length at most  $2c$ ). Then every zone of generation  $k$  has length at most  $(1 - 2\alpha)\rho_k$ , which is less than  $\rho_k$ . In particular, a zone of generation  $k$  that reaches into  $U_k$  must belong to a fraction in the fattened interval, that is, to the dangerous fraction.

**The strategy.** In round  $k$ , Player II looks at Player I's interval  $U_k$  and at the no-go zone of the dangerous fraction of round  $k$ , if there is one. Removing the zone from  $U_k$  leaves at most two intervals, of total length at least  $\rho_k - (1 - 2\alpha)\rho_k = 2\alpha\rho_k$ , so one of them has length at least  $\alpha\rho_k$ . Player II plays her interval  $V_k$  inside that one. That's it.

**Why the strategy wins.** Take any fraction  $p/q$ , and let  $k$  be its generation. The final point  $x$  lies in  $V_k \subseteq U_k$ . If the zone of  $p/q$  reaches into  $U_k$ , then  $p/q$  is the dangerous fraction of round  $k$ , and  $V_k$  avoids its zone, so  $x$  does too. If the zone does not reach into  $U_k$ , then  $x \in U_k$  is outside the zone anyway. Either way  $|x - p/q| \geq c/q^2$ , so  $x \in \mathbf{Bad}(1)$ . The constant  $c$  depends on  $\alpha$ ,  $\beta$  and  $\rho_0$ , which is fine, since the definition of  $\mathbf{Bad}(1)$  allows each number its own constant.

Notice what the game has done for us. The condition “ $|x - p/q| \geq c/q^2$  for all fractions” involves infinitely many fractions at once; the game splits it into infinitely many small jobs, and each job (dodge one interval) is easy.

**Higher dimensions.** For  $\text{Bad}(n)$  the same plan works, with Idea 1 replaced by the *simplex lemma*: if  $n + 1$  rational points in  $\mathbb{R}^n$  with denominators  $q_0, \dots, q_n$  do not lie on a common hyperplane, then the simplex they span has volume at least  $1/(n! q_0 \cdots q_n)$ . (The volume is  $1/n!$  times a determinant whose entries are fractions with denominators  $q_i$ , and a non-zero determinant of that kind is at least  $1/(q_0 \cdots q_n)$  in absolute value, exactly as in Idea 1.) Consequently, all the rational points of one generation that lie in a small ball sit on a single hyperplane, and in each round Player II has to dodge a thin slab around one hyperplane instead of one interval.

**Full dimension.** Suppose Player II has a winning strategy in the  $(\alpha, \beta)$ -game for  $S$ . Inside Player II's ball  $V_k$ , of radius  $\alpha r_k$ , Player I has room for roughly  $\beta^{-n}$  disjoint balls of radius  $\beta \alpha r_k$ , and he may play any of them. Let him try them all, while Player II keeps following her strategy. The result is a tree of plays: every ball has about  $\beta^{-n}$  children, and the radii shrink by the factor  $\alpha\beta$  from one generation to the next. The endpoints of all these plays form a Cantor set, and every point of it lies in  $S$ , because Player II won every one of those plays. A Cantor set built this way has Hausdorff dimension at least about

$$\frac{\log(\beta^{-n})}{\log(1/(\alpha\beta))} = \frac{n \log(1/\beta)}{\log(1/\alpha) + \log(1/\beta)},$$

and this tends to  $n$  as  $\beta \rightarrow 0$ . Since an  $\alpha$ -winning set is winning for *every*  $\beta$ , its dimension is  $n$ . Starting inside any open set gives the local version.

**Countable intersections.** Let  $S_1, S_2, \dots$  be  $\alpha$ -winning. Player II becomes a time-sharer. She splits the rounds into infinitely many infinite arithmetic progressions, and in the rounds of the  $k$ -th progression she plays her strategy for  $S_k$ , ignoring everything else. From the point of view of  $S_k$ , everything that happens between two consecutive rounds of its progression (her moves for the other sets and Player I's moves) just looks like one very big shrinking move by Player I, so the game for  $S_k$  is an  $(\alpha, \beta_k)$ -game with a much smaller  $\beta_k$ . Since  $S_k$  is  $\alpha$ -winning, Player II wins that game too. So the final point lies in every  $S_k$ . Measure and dimension have no such trick, and this is the real reason games are useful here.

## 2. APPLICATIONS TO HOMOGENEOUS DYNAMICS

**The problem.** A *lattice* in  $\mathbb{R}^m$  is a set of the form  $g\mathbb{Z}^m$  with  $g$  an invertible matrix, and it is *unimodular* if  $\det g = \pm 1$ . The unimodular lattices form a space, which can be identified with  $\text{SL}(m, \mathbb{R})/\text{SL}(m, \mathbb{Z})$ . Flows on this space move lattices around by stretching some directions and squeezing others, for example

$$g_t = \text{diag}(e^t, \dots, e^t, e^{-nt}), \quad t \geq 0,$$

acting on  $\mathbb{R}^{n+1}$ . For a flow of this kind almost every orbit is dense, so wanders all over the space, and the lattices whose orbits stay in a bounded region form a set of measure zero. How big is this set of measure zero?

Don't panic. This section uses the words "Lie group", "lattice" and "horospherical subgroup", but the heart of the matter is a computation with  $2 \times 2$  matrices acting on the plane, which we do by hand below. The Lie group language is just the coordinate system that lets the same computation run in higher dimensions. If you understood Section 1, you already know the main idea: this section shows that the game you just played is secretly about lattices being stretched and squeezed.

✂ **2.1. History of the problem and its solution using topological games.** The space of lattices is not compact: you can squash a lattice so that it has a very short non-zero vector, and then squash it further. Mahler’s compactness criterion (1946) says that this is the only way to escape to infinity: a set of unimodular lattices is bounded exactly when the lengths of their shortest non-zero vectors are bounded away from zero. In other words, a lattice can escape to infinity in only one way: by getting squashed. A lattice deep in the cusp is a very thin needle-like grid with a vector of length nearly zero. “Going to infinity” in this space is usually called “going into the cusp”.

The link with the previous section was found by Dani, who showed in 1985 [Dan85] that badly approximable vectors correspond exactly to bounded orbits of the flow above, and in 1986 used Schmidt’s games to show that bounded orbits are abundant for several families of flows [Dan86]. In his plenary address at the International Congress of Mathematicians in Kyoto in 1990 [Mar91], Margulis conjectured that this abundance is universal: for every one-parameter flow that is not quasi-unipotent (roughly, every flow that really stretches something), the set of points with bounded orbits is *thick*, that is, it meets every non-empty open set in a set of full Hausdorff dimension.

The conjecture was proved by Kleinbock and Margulis in 1996 [KM96], by playing Schmidt’s game on the expanding leaves of the flow. Later, Kleinbock and Weiss [KW13] introduced *modified Schmidt games*, adapted to flows that stretch different directions at different rates, and showed in 2013 that sets of bounded orbits are winning for them, which adds the countable intersection property to thickness. Winning has also been established for badly approximable points on planar curves, by An, Beresnevich and Velani in 2018 [ABV18].

*Recognition.* Margulis received the Fields Medal in 1978, the Wolf Prize in 2005 and the Abel Prize in 2020. Dani received the Shanti Swarup Bhatnagar Prize in 1990 and was an invited speaker at the International Congress of Mathematicians in Zürich in 1994. The dynamical approach to Diophantine approximation that grew out of this work led Kleinbock and Margulis to their proof of the Baker–Sprindžuk conjectures, by non-divergence estimates rather than games, in the *Annals of Mathematics* in 1998 [KM98].

**2.2. The Main Result of this section.** The main result consists of Dani’s correspondence and the theorem of Kleinbock and Margulis.

**Theorem 2.1 (Dani, 1985; Kleinbock–Margulis, 1996).**

- (1) Dani’s correspondence. *For  $\mathbf{x} \in \mathbb{R}^n$  let  $\Lambda_{\mathbf{x}} = \begin{pmatrix} I_n & \mathbf{x} \\ 0 & 1 \end{pmatrix} \mathbb{Z}^{n+1}$ . Then  $\mathbf{x} \in \mathbf{Bad}(n)$  if and only if the forward orbit  $\{g_t \Lambda_{\mathbf{x}} : t \geq 0\}$  is bounded.*
- (2) Thickness of bounded orbits. *For every non-quasi-unipotent one-parameter subgroup of a connected semisimple Lie group  $G$  without compact factors, and every lattice  $\Gamma$  in  $G$ , the set of points of  $G/\Gamma$  with bounded orbits is thick.*

Part (1) says that Schmidt’s theorem of Section 1 is secretly a theorem about one particular flow; part (2) says the same phenomenon happens for all flows that stretch.

**2.3. The game.** The game is Schmidt’s game of Section 1, played not on the whole space but on an *expanding leaf* of the flow. The expanding horospherical subgroup  $H$  consists of the elements  $h$  with  $g_{-t} h g_t \rightarrow e$  as  $t \rightarrow \infty$ ; these are the directions that the flow stretches, like pizza dough. Since distances in these directions grow exponentially, a tiny ball soon becomes a ball of unit size, and this is what lets Player II deal with the problem one scale at a time. An orbit  $Hx$  is a copy of a nilpotent Lie group, it carries a natural metric, and Player I and Player II play Schmidt’s game with closed balls in it. Player II wins if the final point  $hx$

has a bounded orbit. For the diagonal flow above,  $H$  consists of the matrices  $\begin{pmatrix} I_n & \mathbf{y} \\ 0 & 1 \end{pmatrix}$ , so the leaf through  $\Lambda_0 = \mathbb{Z}^{n+1}$  is parametrized by  $\mathbf{y} \in \mathbb{R}^n$ , and the game on it is literally Schmidt's game on  $\mathbb{R}^n$ .

**2.4. How the game solves the problem.** Let us see the whole mechanism in the simplest case,  $n = 1$ , where everything can be computed by hand, and then explain what changes in general.

**Lattices in the plane.** For  $n = 1$  we have  $g_t = \text{diag}(e^t, e^{-t})$  and  $u_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$ , so

$$\Lambda_x = u_x \mathbb{Z}^2 = \{(p + qx, q) : p, q \in \mathbb{Z}\}, \quad g_t \Lambda_x = \{(e^t(p + qx), e^{-t}q) : p, q \in \mathbb{Z}\}.$$

The flow stretches the first coordinate and squeezes the second, and the *product* of the two coordinates of a vector,  $q(p + qx)$ , does not change along the orbit. That invariant product is the whole story.

**Dani's correspondence, by hand.** Suppose first that  $x \in \mathbf{Bad}(1)$ , so  $|p + qx| \geq c/|q|$  whenever  $q \neq 0$ . Take a non-zero vector  $(e^t(p + qx), e^{-t}q)$  of  $g_t \Lambda_x$ . If  $q \neq 0$ , the product of its coordinates has absolute value  $|q| |p + qx| \geq c$ , so the two coordinates cannot both be smaller than  $\sqrt{c}$ , and the vector has length at least  $\sqrt{c}$ . If  $q = 0$ , the vector is  $(e^t p, 0)$  with  $p \neq 0$ , of length at least  $e^t \geq 1$ . So every lattice on the orbit has no non-zero vector shorter than  $\min(1, \sqrt{c})$ , and by Mahler's criterion the orbit is bounded.

Conversely, suppose  $x \notin \mathbf{Bad}(1)$ . Then for every  $\varepsilon > 0$  there are  $p$  and  $q \geq 1$  with  $|p + qx| < \varepsilon/q$ . Choose the time  $t$  with  $e^{-t}q = \sqrt{\varepsilon}$ . At that time the vector coming from  $(p, q)$  has second coordinate  $\sqrt{\varepsilon}$  and first coordinate  $e^t |p + qx| < (q/\sqrt{\varepsilon})(\varepsilon/q) = \sqrt{\varepsilon}$ , so it is shorter than  $\sqrt{2\varepsilon}$ . Since  $\varepsilon$  is arbitrary, the orbit dives deeper and deeper into the cusp, and it is unbounded.

So a good rational approximation  $p/q$  of  $x$  is *the same thing* as an excursion of the orbit into the cusp at time  $t \approx \log q$ .

**The dictionary between scales and times.** A one-line computation gives

$$g_t u_x g_{-t} = u_{e^{2t}x},$$

so the flow stretches the expanding leaf  $\{u_x \Lambda : x \in \mathbb{R}\}$  by the factor  $e^{2t}$ . An interval of length  $\rho$  in the leaf is stretched to length about 1 at the time  $t$  with  $e^{2t} \approx 1/\rho$ . Now look again at the proof of Section 1. In round  $k$ , Player I's interval has length  $\rho_k = \rho_0 R^{-k}$ , and the fractions Player II worries about are those of generation  $k$ , with  $q^2 \approx R^k$ ; by the computation above, these are exactly the fractions that send the orbit into the cusp at the time  $t$  with  $e^{2t} \approx R^k \approx 1/\rho_k$ . In other words, *round  $k$  of the game corresponds to the time at which the current interval has been stretched to unit size, and in that round Player II dodges the points whose orbits visit the cusp at that time.* Schmidt's proof, read in this dictionary, is a proof about orbits.

**The general case.** Kleinbock and Margulis play the same game for an arbitrary flow that stretches. Player II works on an expanding leaf  $Hx$ . In each round, she looks at Player I's ball, waits (in her head) until the flow has stretched it to unit size, and asks which points of the ball are deep in the cusp at that time. For the diagonal flow on the plane this bad set was a single short interval, the no-go zone of one fraction. In general there is no single dangerous fraction, and this is where the real work is: Kleinbock and Margulis show, using the mixing properties of the flow, that the bad set occupies only a small proportion of the ball, small enough that Player II can always find a ball of the prescribed size that avoids it. The final point then never goes far into the cusp, and by Mahler's criterion its orbit is bounded. So

the set of points with bounded orbits is winning on every expanding leaf, and hence of full dimension there.

**From leaves to the whole space.** Roughly, near any point the space looks like a product of a piece of an expanding leaf with the directions that the flow does not stretch, and whether an orbit stays bounded depends only mildly on those other directions. So full dimension on every expanding leaf gives full dimension in the whole space, that is, thickness.

### 3. APPLICATIONS TO COMPUTATIONAL COMPLEXITY

**The problem.** A *language* is a set  $L$  of finite binary strings, and a complexity class is a set of languages, such as  $P$  (decidable in polynomial time),  $E = \text{DTIME}(2^{O(n)})$  (decidable in time  $2^{O(n)}$  on inputs of length  $n$ ) or  $\text{ESPACE} = \text{DSpace}(2^{O(n)})$ . We would like to say that “most” languages in  $E$  are hard, or that “most” languages in  $\text{ESPACE}$  need big circuits. Ordinary Baire category and Lebesgue measure on the space of all languages are useless for this: every complexity class is countable, so it is meager and has measure zero, and it looks equally small whether it is  $P$  or  $E$ . The problem is to find a notion of “small” that works *inside* a complexity class.

- ✂ **3.1. History of the problem and its solution using topological games.** The problem was solved by Kurt Mehlhorn in 1973 [Meh73], who introduced resource-bounded Baire category, in which the descriptions of meager sets must themselves be computable with limited resources. In 1990 Jack Lutz reformulated the theory in terms of the Banach–Mazur game played with resource-bounded strategies [Lut90]. He proved, among other results, that languages with high circuit complexity are typical in  $\text{ESPACE}$  in this sense, and in 1992 he introduced the companion notion of resource-bounded measure [Lut92]. These notions became standard tools for quantitative questions about complexity classes.

*Recognition.* Mehlhorn received the Leibniz Prize of the Deutsche Forschungsgemeinschaft in 1986.

**3.2. The Main Result of this section.** The theory rests on the fact that resource-bounded category satisfies a version of the Baire category theorem, so it can tell subclasses of a complexity class apart.

**Theorem 3.1 (Mehlhorn, 1973; Lutz, 1990).**

- (1) Baire category theorem.  $E$  is not meager in itself in the resource-bounded sense.
- (2) The class  $P$  is meager in  $E$  in the resource-bounded sense.
- (3) The class  $\text{SIZE}(2^n/n)$  of languages with circuits of size  $2^n/n$  is meager in  $\text{ESPACE}$  in the resource-bounded sense.

Part (3) says that big circuit complexity is typical in  $\text{ESPACE}$ . Lutz also proved the measure version: in resource-bounded measure, almost every language in  $\text{ESPACE}$  has circuit-size complexity above  $2^n/n$  [Lut92]. Here category and measure agree, although, as on the real line, they can disagree in general.

**3.3. The game.** List all binary strings in the standard order:  $\lambda, 0, 1, 00, 01, 10, 11, 000, \dots$  (shorter strings first, and strings of the same length in dictionary order). A language  $L$  is then the same thing as an infinite 0–1 sequence, its *characteristic sequence*, whose  $N$ -th bit says whether the  $N$ -th string belongs to  $L$ . A finite 0–1 string  $w$  is a partial description of a language: it decides membership for the first  $|w|$  strings. Note that the strings of length  $n$

occupy positions of size about  $2^n$  in this list, so to decide all strings of length  $< n$  we need a description of length about  $2^n$ .

Let  $\mathcal{C}$  be a class of languages.

- (1) The players alternately extend finite binary strings,  $w_0 \sqsubset w_1 \sqsubset w_2 \sqsubset \dots$ , Player I playing  $w_{2k}$  and Player II playing  $w_{2k+1}$ .
- (2) The play produces the language  $L = \bigcup_k w_k$ .
- (3) Player II wins if  $L \notin \mathcal{C}$ .

This is the Banach–Mazur game on the Cantor space, and by the theorem of Banach and Oxtoby Player II has a winning strategy exactly when  $\mathcal{C}$  is meager. The resource-bounded version adds one rule: *Player II is not allowed to think for too long*. Her strategy must compute each move in time polynomial in the length of the current string (or in space polynomial in that length, when we work in ESPACE). A class is *meager in E* if Player II has such a strategy. Player I has no restrictions at all.

**3.4. How the game solves the problem.** The game formulation makes the definition of category effective in the most natural way imaginable: one simply limits the resources of the winning player. The proofs then become constructions of efficient strategies, and they are pleasantly concrete.

**The Baire category theorem: E is not meager in E.** Suppose Player II has a polynomial-time strategy. We show that Player I can make the play produce a language in E, so the strategy does not win the game for the class E. Player I’s strategy is as lazy as possible: at each of his moves he appends a single 0. Now the whole play is computable. To decide whether the  $N$ -th string is in  $L$ , run the game until the current string has length at least  $N$  and read off bit  $N$ . Each round adds at least one bit, so at most  $N$  rounds are needed, and each of Player II’s moves takes time polynomial in the current length, which is at most polynomial in  $N$  (she cannot write more bits than she has time for). So bit  $N$  is computed in time polynomial in  $N$ . Since the strings of length  $n$  sit at positions  $N \approx 2^n$ , membership of a string of length  $n$  is decided in time polynomial in  $2^n$ , that is, in time  $2^{O(n)}$ . So  $L \in E$ .

**P is meager in E: diagonalization, one move at a time.** Let  $M_1, M_2, \dots$  be a list of Turing machines in which  $M_i$  halts within  $n^i + i$  steps on inputs of length  $n$ , so that every language in P is decided by some  $M_i$ . Player II wants to make sure that  $L$  disagrees with every  $M_i$  on at least one string. Here is her move. Suppose the current string  $w$  decides all strings of length less than  $s$ , so  $|w|$  is about  $2^s$ . For each  $i \leq \log s$ , Player II picks a fresh string  $x_i$  of length  $s$  that  $w$  has not decided yet, runs  $M_i$  on  $x_i$ , and extends  $w$  so that

$$x_i \in L \iff M_i \text{ rejects } x_i,$$

filling all the other new positions with 0. Each simulation takes at most  $s^{\log s} + \log s$  steps, which is less than  $2^s$  once  $s$  is large, so the whole move takes time polynomial in  $2^s \approx |w|$ , as the rules require. Player I may then extend the string as much as he likes, but he cannot undo what she wrote. Since Player II moves infinitely often and  $s \rightarrow \infty$ , every  $M_i$  is eventually defeated on some string, so  $L \notin P$ .

The one clever point is the restriction  $i \leq \log s$ . Player II cannot afford to simulate  $M_i$  on short strings when  $i$  is large, since  $n^i$  steps would blow her time budget, so she politely waits until the strings are long enough that  $M_i$  is cheap to simulate relative to  $|w|$ . For instance, if she tried to defeat  $M_{1000}$  in her very first move, the  $n^{1000}$  steps of the simulation would bankrupt her polynomial time budget on the spot. So she waits until  $s \geq 2^{1000}$ , when simulating  $M_{1000}$  is pocket change. This “diagonalization with delay” is a standard trick of complexity theory, and the game makes it look natural.

**Circuit size: a counting argument, one length at a time.** Here the resource bound is space instead of time, and the key fact is Shannon’s counting argument (1949). There are  $2^{2^n}$  Boolean functions of  $n$  variables, one for each truth table. A circuit with  $s$  gates can be described by listing, for each gate, its operation and its two inputs, and a careful count (taking into account that renaming the gates does not change the circuit) shows that for large  $n$  there are fewer circuits of size  $2^n/n$  than there are truth tables. So some Boolean function of  $n$  variables has no circuit of size  $2^n/n$ .

Player II uses this as follows. At her move, she takes the smallest length  $n$  such that no string of length  $n$  has been decided yet. She finds the first truth table on  $n$  variables that has no circuit of size  $2^n/n$ , by brute force: go through the truth tables one by one, and for each one go through all circuits of size  $2^n/n$  and check whether any of them computes it. Then she defines  $L$  on the  $2^n$  strings of length  $n$  to be that truth table. The search takes an astronomical amount of *time*, but very little *space*: at any moment she only needs to remember one truth table and one circuit, which takes space polynomial in  $2^n$ , hence polynomial in the length of the current string. Space can be reused; time cannot. Player II’s search takes longer than the age of the universe, but she is only charged for the scratch paper. Since this happens for infinitely many  $n$ , the final language has no circuits of size  $2^n/n$  at infinitely many lengths, so  $L \notin \text{SIZE}(2^n/n)$ .

#### 4. APPLICATIONS TO TOPOLOGICAL GROUPS

**The problem.** A *semitopological group* is a group with a topology in which multiplication is *separately* continuous: the translations  $x \mapsto xa$  and  $x \mapsto ax$  are continuous. It is a *topological group* if multiplication is jointly continuous and inversion is continuous. Separate continuity is much weaker than joint continuity (think of  $f(x, y) = xy/(x^2 + y^2)$  on the plane, which is continuous in each variable separately but not at the origin). When does separate continuity of multiplication force a semitopological group to be a topological group? Behind this lies a more general question: when is a separately continuous function  $f: X \times Y \rightarrow \mathbb{R}$  jointly continuous at many points?

✂ 4.1. **History of the problem and its solution using topological games.** Baire proved in his thesis of 1899 that a separately continuous function on  $\mathbb{R}^2$  is jointly continuous at the points of a dense  $G_\delta$  set. In 1957 Ellis proved that every locally compact Hausdorff semitopological group is a topological group [Ell57], and in 1974 Namioka proved a general joint continuity theorem for functions on  $X \times K$ , with  $X$  complete in a suitable sense and  $K$  compact, from which he deduced Ellis’s theorem [Nam74].

The game-theoretic solution came in three steps. In 1981 Christensen replaced the completeness hypothesis on  $X$  in Namioka’s theorem by the existence of a winning strategy in a topological game [Chr81]. In 1983 Saint-Raymond showed that it is enough that the opponent have no winning strategy, and deduced that every metrizable Baire space has the joint continuity property [SR83]. For groups, Bouziad proved in 1996 that every Čech-analytic Baire semitopological group is a topological group [Bou96], and in 2001 Kenderov, Kortezov and Moors introduced a game that describes a class of spaces, the strongly Baire spaces, on which every semitopological group is a topological group [KKM01].

4.2. **The Main Result of this section.** The main result consists of Namioka’s theorem, Ellis’s theorem, and the theorem of Kenderov, Kortezov and Moors, which contains Ellis’s.

**Theorem 4.1 (Ellis, 1957; Namioka, 1974; Kenderov–Kortezov–Moors, 2001).**

- (1) Namioka’s theorem. *Let  $X$  be a Čech-complete space,  $K$  a compact space, and  $f: X \times K \rightarrow \mathbb{R}$  separately continuous. Then there is a dense  $G_\delta$  set  $A \subseteq X$  such that  $f$  is jointly continuous at every point of  $A \times K$ .*
- (2) Ellis’s theorem. *Every locally compact Hausdorff semitopological group is a topological group.*
- (3) *Every semitopological group whose underlying space is strongly Baire is a topological group.*

Every regular Čech-complete space, in particular every locally compact Hausdorff space and every complete metric space, is strongly Baire, so part (3) contains part (2). Readers of our notes on stability will recognize the theme:  $f$  is jointly continuous at every point of  $\{x\} \times K$  exactly when the map  $x \mapsto f(x, \cdot)$  from  $X$  to  $C(K)$ , which is continuous for the topology of pointwise convergence, is continuous at  $x$  for the norm topology. So Namioka’s theorem is about upgrading pointwise convergence to uniform convergence, at least at many points.

**4.3. The game.** Let  $X$  be a space and  $D \subseteq X$  a dense set. In the game  $G_S(D)$  of Kenderov, Kortezov and Moors, their  $\beta$  is our Player I and their  $\alpha$  is our Player II.

- (1) Player I chooses a non-empty open set  $U_1 \subseteq X$ .
- (2) Player II chooses a non-empty open set  $V_1 \subseteq U_1$ .
- (3) In general, Player I chooses a non-empty open  $U_{k+1} \subseteq V_k$  and Player II a non-empty open  $V_{k+1} \subseteq U_{k+1}$ .
- (4) Player II wins if  $\bigcap_k V_k \neq \emptyset$  and every sequence  $(a_k)$  with  $a_k \in V_k \cap D$  has a cluster point in  $X$ .

A regular space is *strongly Baire* if, for some dense  $D$ , Player I has no winning strategy. This is the Banach–Mazur game with a more demanding winning condition for Player II; the games of Christensen and Saint-Raymond are variants in which Player II also picks points as she goes.

To see an example, let  $X$  be Čech-complete, say  $X = \bigcap_k G_k$  with  $G_k$  open in a compactification  $bX$ . Player II wins with  $D = X$  by choosing  $V_k$  whose closure in  $bX$  lies inside  $G_k \cap U_k$ . The closures form a decreasing sequence of non-empty compact sets, their intersection lies in  $X$  and equals  $\bigcap_k V_k$ , and every sequence with  $a_k \in V_k$  clusters in it.

**4.4. How the game solves the problem.** The arguments of Christensen, Saint-Raymond and Kenderov–Kortezov–Moors share a pattern that is worth learning, because it is a little like a detective story. Suppose, for instance, that multiplication is not jointly continuous at  $(e, e)$ . Then there is a neighbourhood  $W$  of  $e$  such that every pair of neighbourhoods of  $e$  contains elements  $u, v$  with  $uv \notin W$ . These pairs are the clues. Player I plays the Accuser, trying to prove that multiplication is guilty of discontinuity, and Player II plays the Defender. The Accuser uses the clues to build a strategy: at each move he chooses an open set inside Player II’s last move that records a new pair of witnesses, chosen so that separate continuity at any cluster point of the witnesses would be violated. Since  $X$  is strongly Baire, this strategy cannot be a winning one, so some play against it is won by Player II. That play hands us a cluster point of the witnesses, and separate continuity at that point gives a contradiction. In short, the failure of joint continuity is turned into a strategy for Player I, and the topological hypothesis says exactly that such strategies can be beaten.

Continuity of inversion is handled separately. For locally compact groups this gives Ellis's theorem; Namioka's original deduction, which uses translations to spread a single point of joint continuity over the whole group, follows the same idea.

## 5. APPLICATIONS TO BANACH SPACE THEORY

**The problems.** (1) *Differentiability of convex functions.* In calculus, a convex function of one variable is differentiable at all but countably many points. In infinite dimensions things are much less tidy. A Banach space  $X$  is an *Asplund space* if every continuous convex function on an open convex subset of  $X$  is Fréchet differentiable on a dense  $G_\delta$  set. Which spaces are Asplund spaces?

(2) *The unconditional basic sequence problem.* Does every infinite-dimensional Banach space contain an unconditional basic sequence, that is, a sequence  $(e_n)$  that is a basis of the subspace it spans and for which every convergent expansion  $\sum a_n e_n$  keeps converging when you flip the signs of the terms or shuffle their order?

✂ **5.1. History of the problems and their solutions using topological games.** Mazur proved in 1933 that continuous convex functions on separable Banach spaces are Gâteaux differentiable on dense  $G_\delta$  sets, and Asplund introduced the spaces now named after him in 1968, in a paper in *Acta Mathematica* [Asp68]. The characterization problem was solved in 1975 by Namioka and Phelps [NP75] and by Stegall [Ste75]. The game-theoretic description of the property behind it, fragmentability, was found by Kenderov and Moors and published in 1999 [KM99].

The unconditional basic sequence problem was one of the central open problems of Banach space theory for decades. It was solved, negatively, by Gowers and Maurey in 1992 (published in 1993) [GM93]. They built a space so tangled that no subspace splits as a direct sum of two infinite-dimensional subspaces, a *hereditarily indecomposable* space: a Banach space so tangled that not even its subspaces can be split up amicably. Gowers then showed, in 1996 and 2002, using an infinite game played with subspaces and vectors, that such strange spaces are not an accident: every Banach space contains either a subspace with an unconditional basis or a hereditarily indecomposable subspace [Gow96, Gow02].

*Recognition.* Gowers received the Fields Medal in 1998, for research connecting functional analysis and combinatorics, which includes the solution of the unconditional basic sequence problem and the dichotomy theorem. He was an invited speaker at the International Congress of Mathematicians in Zürich in 1994 and received the Prize of the European Mathematical Society in 1996. Asplund's paper appeared in *Acta Mathematica*, and Gowers's Ramsey theorem in the *Annals of Mathematics* in 2002.

**5.2. The Main Results of this section.** Why should differentiability have anything to do with the dual space? The derivative of a function on  $X$  at a point is a linear functional, an element of  $X^*$ , so differentiability is a question about how the candidate derivatives (the subgradients) sit inside  $X^*$ . Roughly, Fréchet differentiability at a point means that subgradients at nearby points are close to the derivative *in norm*, and the norm is what fragmentability controls. The first result collects several characterizations of Asplund spaces; one of them is exactly what the game of Kenderov and Moors detects. The second is Gowers's dichotomy.

**Theorem 5.1 (Asplund, 1968; Namioka–Phelps, 1975; Stegall, 1975).** *For a Banach space  $X$  the following are equivalent:*

- (1)  $X$  is an Asplund space;
- (2) every separable subspace of  $X$  has a separable dual;
- (3)  $X^*$  has the Radon–Nikodým property;
- (4) the dual ball  $B_{X^*}$ , with the weak\* topology, is fragmented by the norm: every non-empty subset of  $B_{X^*}$  has non-empty relatively weak\*-open subsets of arbitrarily small norm diameter.

Kenderov and Moors [KM99] showed that a space  $(K, \tau_1)$  is fragmented by a metric whose topology contains  $\tau_2$  if and only if Player II has a winning strategy in the game described below. In particular, if  $X$  is an Asplund space, Player II wins that game on  $B_{X^*}$  with the weak\* and norm topologies.

**Theorem 5.2 (Gowers’s dichotomy, 1996 and 2002).** *Every infinite-dimensional Banach space contains an infinite-dimensional subspace that either has an unconditional basis or is hereditarily indecomposable.*

The two alternatives cannot happen at the same time, because a hereditarily indecomposable space contains no unconditional basic sequence. So every Banach space, however large, contains a subspace that is either perfectly civilized or thoroughly rebellious; there is no third option.

### 5.3. The games.

5.3.1. *The fragmentability game.* Let  $K$  be a set with two topologies  $\tau_1$  and  $\tau_2$ ; for the dual ball,  $\tau_1$  is the weak\* topology and  $\tau_2$  the norm topology.

- (1) Player I chooses a non-empty subset  $U_0 \subseteq K$ .
- (2) Player II chooses a non-empty relatively  $\tau_1$ -open subset  $V_0 \subseteq U_0$ .
- (3) In round  $k \geq 1$ , Player I chooses a non-empty subset  $U_k \subseteq V_{k-1}$ , and Player II a non-empty relatively  $\tau_1$ -open subset  $V_k \subseteq U_k$ .
- (4) Player II wins if  $\bigcap_k U_k$  is empty, or if it is a single point  $x$  and every  $\tau_2$ -neighbourhood of  $x$  contains some  $U_k$ .

This is a Banach–Mazur game with a twist: Player I may play any non-empty set at all, while Player II must play relatively open ones, and she wins if the sets shrink down to a point in the stronger topology.

5.3.2. *Gowers’s game.* Let  $X$  have a basis, and let  $\Sigma$  be a set of sequences of normalized block vectors. In round  $k \geq 1$ , Player I chooses an infinite-dimensional block subspace  $Y_k$ , and Player II chooses a normalized block vector  $x_k \in Y_k$ . Player II wins if  $(x_1, x_2, \dots) \in \Sigma$ . Player I’s subspaces play the role of open sets. A block subspace can be chosen to live as far out along the basis as Player I likes, so by choosing it he can force Player II’s next vector to be supported beyond any finite stage, and he can restrict the kind of vectors available to her. That is exactly what a small neighbourhood does in the Banach–Mazur game: it limits where the other player may look.

### 5.4. How the games solve the problems.

5.4.1. *Asplund spaces.* If  $K$  is fragmented by the norm, Player II has an easy strategy: answer each move  $U_k$  with a relatively weak\*-open subset  $V_k \subseteq U_k$  of norm diameter less than  $2^{-k}$ . Then  $\bigcap_k U_k$  has norm diameter 0, so it is empty or a single point  $x$ , and in the second case  $U_{k+1} \subseteq V_k$  lies in the norm ball of radius  $2^{-k}$  about  $x$ . Conversely, Kenderov and Moors build a fragmenting metric out of a winning strategy for Player II; roughly, two points are close when the strategy takes a long time to separate them. The advantage of the game is that a strategy is a local object, built one move at a time, while a fragmenting metric is a global one, and it is often much easier to describe how to play than to write down a metric.

5.4.2. *Gowers's dichotomy.* Gowers's infinite Ramsey theorem [Gow02] says that for every analytic set  $\Sigma$  and every sequence of positive tolerances, there is a block subspace  $Y$  such that either no block sequence of  $Y$  lies in  $\Sigma$ , or Player II has a winning strategy in the game played in  $Y$  for a slight enlargement of  $\Sigma$ , obtained by wiggling its members within the given tolerances. This is a game-theoretic version of the Galvin–Prikry and Ellentuck theorems on Ramsey sets (Section 6), adapted to Banach spaces. An exact Ramsey statement is false for vectors, and the game is what can be rescued. The proof uses a “combinatorial forcing” argument, in which subspaces are said to accept or reject finite block sequences.

Roughly, the dichotomy follows by applying the Ramsey theorem to the set of block sequences that fail to be  $C$ -unconditional. If some block subspace avoids these sequences for some constant  $C$ , every block basis in it is  $C$ -unconditional, and it has an unconditional basis. Otherwise, in every block subspace Player II can produce badly unconditional block sequences, and from her strategies Gowers extracts a subspace in which no two infinite-dimensional subspaces form a direct sum, that is, a hereditarily indecomposable subspace. The game is what locates this subspace; the existence of such spaces is the theorem of Gowers and Maurey.

## 6. APPLICATIONS TO INFINITE-DIMENSIONAL RAMSEY THEORY

**The problem.** Ramsey's theorem (1930) says that if you colour the  $n$ -element subsets of  $\omega = \{0, 1, 2, \dots\}$  with two colours, there is an infinite  $H \subseteq \omega$  all of whose  $n$ -element subsets get the same colour. What if we colour the *infinite* subsets instead? Call  $\mathcal{A} \subseteq [\omega]^\omega$ , the set of infinite subsets of  $\omega$ , *Ramsey* if there is an infinite  $H \subseteq \omega$  such that  $[H]^\omega \subseteq \mathcal{A}$  or  $[H]^\omega \cap \mathcal{A} = \emptyset$ . Using the axiom of choice one can build sets that are not Ramsey. Which sets are Ramsey?

✂ 6.1. **History of the problem and its solution using topological games.** Ramsey's theorem appeared in 1930 [Ram30], and in 1952 Erdős and Rado showed, using the axiom of choice, that its infinite-dimensional analogue fails [ER52]. Nash-Williams proved in 1965 that clopen sets, in the topology inherited from the Cantor space, are Ramsey [NW65]; Galvin and Prikry extended this to Borel sets (published in 1973) [GP73], and Silver to analytic sets (published in 1970) [Sil70], by a proof that used metamathematical methods. Mathias developed the theory further, with applications to forcing, in his work on happy families [Mat77].

The problem was solved by Erik Ellentuck in 1974 [Ell74], who introduced a new topology on  $[\omega]^\omega$ , finer than the usual one, and proved that the sets that are Ramsey in a hereditary sense are exactly those with the Baire property in this topology. This gave a short proof of Silver's theorem. In 1983 Kastanas [Kas83] characterized the same sets by the determinacy of a game, which is a Banach–Mazur game in Ellentuck's topology. Gowers's game in Section 5 is the analogue of Kastanas's game for block sequences in Banach spaces.

**6.2. The Main Result of this section.** The open sets in this section look strange at first, so here is the idea. In the usual (Cantor space) topology on  $[\omega]^\omega$ , two infinite sets are close if they begin in the same way, that is, if they share the same finite initial segment  $s$ . Ellentuck's open sets ask for more: they fix the beginning  $s$  *and* restrict the rest of the set to come from a given reservoir  $M$ . For a finite  $s \subseteq \omega$  and an infinite  $M \subseteq \omega$  with  $\max s < \min M$ , let

$$[s, M] = \{X \in [\omega]^\omega : s \text{ is an initial segment of } X \text{ and } X \subseteq s \cup M\}.$$

Think of  $s$  as a stem that has already been decided and of  $M$  as the pool of candidates for the rest. These sets form a basis for the *Ellentuck topology*. A set  $\mathcal{A}$  is *completely Ramsey* if for every  $[s, M]$  there is an infinite  $N \subseteq M$  with  $[s, N] \subseteq \mathcal{A}$  or  $[s, N] \cap \mathcal{A} = \emptyset$ .

**Theorem 6.1 (Ellentuck, 1974; Kastanas, 1983).** *For  $\mathcal{A} \subseteq [\omega]^\omega$  the following are equivalent:*

- (1)  $\mathcal{A}$  is completely Ramsey;
- (2)  $\mathcal{A}$  has the Baire property in the Ellentuck topology;
- (3) for every  $s$  and  $N$ , the Kastanas game  $K(\mathcal{A}, s, N)$  described below is determined.

Analytic sets have the Baire property in every topology finer than the one in which they are analytic, so every analytic set, and in particular every Borel set, is completely Ramsey. This recovers the theorems of Galvin–Prikry and Silver in one stroke.

**6.3. The game.** The Kastanas game  $K(\mathcal{A}, s, N)$  is played as follows.

- (1) Player I chooses an infinite  $M_0 \subseteq N$ .
- (2) Player II chooses  $n_0 \in M_0$  and an infinite  $N_0 \subseteq M_0$  with  $n_0 < \min N_0$ .
- (3) In general, Player I chooses an infinite  $M_k \subseteq N_{k-1}$ , and Player II chooses  $n_k \in M_k$  and an infinite  $N_k \subseteq M_k$  with  $n_k < \min N_k$ .
- (4) Player II wins if  $s \cup \{n_0, n_1, \dots\} \in \mathcal{A}$ .

In Kastanas's paper the payoff set is Player I's; since the theorem is about determinacy and  $\mathcal{A}$  is completely Ramsey exactly when its complement is, the switch does not matter. The moves are basic open sets in disguise: Player I's move  $M_k$  stands for  $[s \cup \{n_0, \dots, n_{k-1}\}, M_k]$ , and Player II answers with the smaller basic open set  $[s \cup \{n_0, \dots, n_k\}, N_k]$ . So this is a Banach–Mazur game in the Ellentuck topology, in which Player II must commit to one more element at each move.

**6.4. How the game solves the problem.** All three theorems rest on the same device, the *combinatorial forcing* of Galvin and Prikry. Say that  $M$  *accepts*  $s$  if  $[s, M] \subseteq \mathcal{A}$ , and that  $M$  *rejects*  $s$  if no infinite  $N \subseteq M$  accepts  $s$ . In plain words,  $M$  accepts the stem  $s$  if, however you complete  $s$  with infinitely many elements of  $M$ , you can never escape from  $\mathcal{A}$ ; and  $M$  rejects  $s$  if no amount of thinning out  $M$  will make it accept. By a diagonal argument one finds, inside any  $M$ , an infinite  $N$  that has made up its mind about every finite  $s \subseteq N$ : it accepts or rejects each of them, and the decisions fit together. Ellentuck's insight was that these diagonal arguments are exactly the proof that the Ellentuck space is a Baire space, and that the sets with  $[s, N] \cap \mathcal{A} = \emptyset$  inside every  $[s, M]$  are exactly the nowhere dense sets. Kastanas's game turns the diagonalization into a play: a winning strategy for Player II, run against all possible moves of Player I and glued together by a diagonal fusion, produces an infinite  $M \subseteq N$  with  $[s, M] \subseteq \mathcal{A}$ , and a winning strategy for Player I produces one with  $[s, M] \cap \mathcal{A} = \emptyset$ . So the Ramsey property is to Kastanas's game what the Baire property is to the Banach–Mazur game.

## Part II. Builders: other topological games

In Part I the players were sculptors, chipping away at a set until only a point was left. In Part II they are builders: they choose covers, call out numbers, and compare sets with one another.

### 7. APPLICATIONS TO SET-THEORETIC TOPOLOGY

**The problem.** Compact sets are the “small” sets of topology, and countable unions of compact sets ( $\sigma$ -compact sets) are almost as tame. In 1924 Karl Menger proposed a covering property that looked like a clean characterization of  $\sigma$ -compactness. A space  $X$  has the *Menger property* if for every sequence  $\mathcal{U}_1, \mathcal{U}_2, \dots$  of open covers of  $X$  there are finite sets  $\mathcal{F}_n \subseteq \mathcal{U}_n$  such that  $\bigcup_n \mathcal{F}_n$  covers  $X$ . Every  $\sigma$ -compact space has the Menger property: if  $X = \bigcup_n K_n$  with  $K_n$  compact, choose  $\mathcal{F}_n$  to be a finite subcover of  $K_n$ . Menger conjectured the converse for metric spaces. The problem is to decide Menger’s conjecture, in particular for sets of real numbers.

✂ **7.1. History of the problem and its solution using topological games.** Menger introduced the property in 1924 [Men24], in a different but equivalent form, while studying a basis property of metric spaces, and conjectured that it characterizes  $\sigma$ -compactness. Hurewicz gave it the form above in 1925 [Hur26], characterized it by an infinite game, and introduced a stronger property for which he made the analogous conjecture. Under the continuum hypothesis the conjecture fails: a Luzin set, an uncountable set of reals meeting every meager set in a countable set, has the Menger property but is not  $\sigma$ -compact. In 1975 Telgársky characterized  $\sigma$ -compactness by the same game [Tel75], so that Menger’s conjecture became the statement that the Menger game on a set of reals is always determined.

The problem was solved by David Fremlin and Arnold Miller in 1988 [FM88], who constructed, in ZFC, a set of reals with the Menger property that is not  $\sigma$ -compact. The closely related property of Rothberger (1938), in which a single member of each cover is chosen, has a parallel game-theoretic story: Galvin introduced the point-open game in 1978 [Gal78], and in 1994 Pawlikowski proved that a set of reals is undetermined in this game if and only if it is uncountable and has Rothberger’s property [Paw94]. Borel conjectured in 1919 that every set of reals of strong measure zero is countable, and Laver proved in 1976 that this is consistent [Lav76]. Since Rothberger’s property implies strong measure zero, in Laver’s model every set of reals is determined in the point-open game, while under the continuum hypothesis a Luzin set gives an undetermined one [Rec94].

**7.2. The Main Result of this section.** The main result consists of the game characterizations of the Menger property and of  $\sigma$ -compactness, and the theorem of Fremlin and Miller.

**Theorem 7.1 (Hurewicz, 1925; Telgársky, 1975; Fremlin–Miller, 1988).**

- (1) *A space  $X$  has the Menger property if and only if Player I has no winning strategy in the Menger game on  $X$ .*
- (2) *A separable metrizable space  $X$  is  $\sigma$ -compact if and only if Player II has a winning strategy in the Menger game on  $X$ .*
- (3) *There is a set of reals with the Menger property that is not  $\sigma$ -compact. Consequently, the Menger game on this set is not determined, and Menger’s conjecture is false.*

**7.3. The games.** In the *Menger game* on  $X$ , in round  $n$  Player I opens an umbrella: he chooses an open cover  $\mathcal{U}_n$  of the whole space  $X$ , and Player II chooses a finite set  $\mathcal{F}_n \subseteq \mathcal{U}_n$ . Player II wins if  $\bigcup_n \mathcal{F}_n$  covers  $X$ . In the *Rothberger game* Player II must be even more frugal and chooses a single member of each cover. In the literature Player I is called ONE and Player II is called TWO. In Galvin's *point-open game*, Player I chooses a point  $x_n \in X$  and Player II an open set  $U_n \ni x_n$ , and Player II wins if the sets  $U_n$  do not cover  $X$ ; it is dual to the Rothberger game, in the sense that a winning strategy for either player in one game corresponds to a winning strategy for the other player in the other game. Watch out for the reversal: in the Menger and Rothberger games Player II wants her umbrellas to keep all of  $X$  dry, while in Galvin's game she places the umbrellas but wants to leave some spot out in the rain.

#### 7.4. How the games solve the problem.

**Menger's property.** If Player I has no winning strategy, then, given covers  $\mathcal{U}_1, \mathcal{U}_2, \dots$ , the strategy "play  $\mathcal{U}_n$  in round  $n$ " is not winning, so some play against it is won by Player II, and her moves witness the Menger property. The converse is the substantial direction: a strategy for Player I produces only countably many covers, one for each finite position, since a Lindelöf space allows Player I's covers to be taken countable, and Hurewicz applies the Menger property to a suitable arrangement of these covers to build a play that defeats the strategy.

**$\sigma$ -compactness.** If  $X = \bigcup_n K_n$  with  $K_n$  compact and increasing, Player II wins by choosing, in round  $n$ , a finite subfamily of  $\mathcal{U}_n$  that covers  $K_n$ . Conversely, Telgársky unravels a winning strategy of Player II, position by position, into countably many compact sets that cover  $X$ .

**Menger's conjecture.** By parts (1) and (2), a separable metrizable space with the Menger property that is not  $\sigma$ -compact is exactly one on which neither player has a winning strategy. Fremlin and Miller construct such a set of reals from combinatorial properties of families of functions in  $\omega^\omega$  under eventual domination; the link is a theorem of Hurewicz, which says that a subset of  $\omega^\omega$  has the Menger property exactly when none of its continuous images in  $\omega^\omega$  is a dominating family. The point of view of games explains both why the conjecture is natural and why it fails. It is natural because it asserts a form of determinacy, and determinacy holds for well-behaved sets (for analytic sets of reals the conjecture is in fact true). It fails because games played on sufficiently wild sets need not be determined, and Fremlin and Miller found a wild enough set without any extra axioms.

## 8. APPLICATIONS TO DESCRIPTIVE SET THEORY

**The problem.** Here is the simplest infinite game imaginable. Fix a set  $A \subseteq \omega^\omega$  of infinite sequences of natural numbers. In the *Gale-Stewart game*  $G(A)$ , Player I and Player II take turns calling out natural numbers, Player I playing  $x(0), x(2), \dots$  and Player II playing  $x(1), x(3), \dots$ , and Player II wins if the resulting sequence  $x$  belongs to  $A$ . The game is *determined* if one of the players has a winning strategy. For finite games this is always the case (that is Zermelo's theorem, and it is why chess has an optimal strategy that nobody knows). For infinite games it can fail: using the axiom of choice one can build sets  $A$  for which neither player has a winning strategy. Which sets, and in particular which Borel sets, give determined games? And what does the answer tell us about the structure of the Borel sets?

- ✂ 8.1. **History of the problem and its solution using infinite games.** The story of this problem is unusual: the game came first, and the applications came later. Gale and Stewart

introduced these games in 1953 [GS53], proved that  $G(A)$  is determined when  $A$  is open or closed, constructed an undetermined game from a well-ordering of the reals, and asked how far determinacy extends. Wolfe proved in 1955 that games with  $F_\sigma$  or  $G_\delta$  payoff sets are determined [Wol55], and Davis extended this in 1964 to the next level of the Borel hierarchy [Dav64]. In 1970 Martin proved that analytic games are determined if there is a measurable cardinal [Mar70], and in 1971 Friedman showed that any proof of Borel determinacy must use uncountably many iterations of the power set operation [Fri71], so that the problem could not be solved by the methods of ordinary analysis. A statement about sets of real numbers that needs sets of sets of sets of  $\dots$  of reals, iterated uncountably many times, for its proof: that was something new. Martin's argument may well be the most expensive proof ever written of a statement about sets of real numbers.

The problem was solved by Donald A. Martin in 1975 [Mar75], who proved that every Borel game is determined. Even before this, Wadge had shown in his work of 1972, written up in his thesis of 1983 [Wad83], that determinacy turns the Borel sets into a hierarchy that is essentially a well-ordering, by comparing sets through a game of his own. The theory was later extended far beyond the Borel sets: Martin and Steel proved in 1989 that projective games are determined if there are infinitely many Woodin cardinals [MS89].

*Recognition.* Martin's proof appeared in the *Annals of Mathematics*. Martin, Steel and Woodin received the Karp Prize of the Association for Symbolic Logic in 1988 for their work on determinacy, and the proof of projective determinacy by Martin and Steel appeared in the *Journal of the American Mathematical Society*.

**8.2. The Main Result of this section.** The main result consists of Martin's theorem and its consequence for the comparison of Borel sets. For  $A, B \subseteq \omega^\omega$  write  $A \leq_W B$ , and say that  $A$  is *Wadge reducible* to  $B$ , if there is a continuous  $f: \omega^\omega \rightarrow \omega^\omega$  with  $A = f^{-1}(B)$ .

**Theorem 8.1 (Martin, 1975; Wadge, 1972).**

- (1) Borel determinacy. *For every Borel set  $A \subseteq \omega^\omega$ , the game  $G(A)$  is determined.*
- (2) Wadge's lemma. *For all Borel sets  $A, B \subseteq \omega^\omega$ , either  $A \leq_W B$  or  $B \leq_W \omega^\omega \setminus A$ .*

By part (2), any two Borel sets are comparable under  $\leq_W$ , up to complements. Martin and Monk showed that  $\leq_W$  is also well-founded on the Borel sets, so the Borel sets, up to Wadge equivalence, form a hierarchy indexed by ordinals that refines the classical Borel hierarchy.

**8.3. The games.** The Gale–Stewart game was described in the box above. The *Wadge game*  $W(A, B)$  is a clever way of comparing two sets. Think of Player II as trying to imitate Player I: whenever his sequence lands in  $A$ , hers should land in  $B$ , and vice versa. Player I plays  $x(0), x(1), \dots$  and Player II plays  $y(0), y(1), \dots$ , alternately, Player I first, and Player II wins if

$$x \in A \iff y \in B.$$

This is a Gale–Stewart game whose payoff set is Borel when  $A$  and  $B$  are.

**8.4. How the games solve the problem.**

**Open games.** Suppose that  $A$  is open and that Player II has no winning strategy in  $G(A)$ . Call a finite position *good for Player I* if Player II has no winning strategy from it. If a position with Player I to move is good for him, some move keeps it good, for otherwise Player II would have a winning strategy from every successor position and hence from the position itself; if a position with Player II to move is good for Player I, every move keeps it good. So Player I can play forever through good positions. If the resulting  $x$  were in  $A$ , then,

$A$  being open, some initial segment of  $x$  would already force the play into  $A$ , and Player II would win trivially from that position, which would not be good for Player I. Hence  $x \notin A$ , and Player I has a winning strategy. The idea is simply “never step into a position from which you are already lost”; the openness of  $A$  guarantees that this cautious play is enough.

**Borel games.** Martin’s proof reduces every Borel game to an open game on a larger tree. Roughly, for each Borel set  $A$  he builds a game  $G'$  whose moves are more complicated objects than natural numbers, together with a map from plays of  $G'$  to plays of  $G(A)$  that sends strategies to strategies, such that the payoff set of  $G'$  is clopen and corresponds to  $A$ . Since  $G'$  is determined by the Gale–Stewart theorem, so is  $G(A)$ . The construction proceeds by induction on the Borel rank of  $A$ , and at each step the moves of the new game are sets of positions of the old one; this is why the proof climbs through the iterated power sets of  $\mathbb{R}$ , in agreement with Friedman’s theorem.

**Wadge’s lemma.** By Borel determinacy, one of the players has a winning strategy in  $W(A, B)$ . If Player II has one, her strategy is literally a blueprint for a continuous function: Player I feeds in an input sequence  $x$  one digit at a time, and Player II’s step-by-step answers are the output  $y$ . Since each output digit depends only on finitely many input digits, the map  $x \mapsto y$  is continuous, and  $x \in A$  if and only if  $y \in B$ , so  $A \leq_W B$ . If Player I has one, his strategy turns each sequence  $y$  of moves of Player II into a sequence  $x$  continuously, with  $y \in B$  if and only if  $x \notin A$ , so  $B \leq_W \omega^\omega \setminus A$ . The game thus produces the reduction itself, and the comparability of Borel sets is literally the determinacy of a game.

## Appendix: Applications of non-topological games

The games in this appendix are not played with open sets at all: they are finite games on boards, or games played with elements of two structures. They are included because they are too beautiful to leave out, and because they show the same two features of games (local decisions, combinable strategies) at work far from topology.

### APPENDIX A. APPLICATIONS TO THE PROBABILISTIC METHOD

**The problem.** The probabilistic method proves that combinatorial objects exist by showing that a random object has the required property with positive probability. For example, Erdős proved in 1963 that if a family  $\mathcal{F}$  of  $n$ -element sets has fewer than  $2^{n-1}$  members, then the elements can be coloured red and blue so that no member of  $\mathcal{F}$  is monochromatic: a random colouring makes a given set monochromatic with probability  $2^{1-n}$ , so the expected number of monochromatic sets is less than 1. Such proofs give no construction. The problem is to turn them into explicit, deterministic, efficient constructions.

- ✂ **A.1. History of the problem and its solution using combinatorial games.** The probabilistic method was introduced by Erdős in 1947, in his lower bound for diagonal Ramsey numbers [Erd47], and the colouring theorem above is from 1963 [Erd63]. In 1973 Erdős and Selfridge studied Maker–Breaker games, in which the players alternately claim elements of a board and Maker tries to claim all the elements of some winning set [ES73]. They gave a strategy for Breaker based on a potential function equal to the expected number of winning sets that Maker would complete if the rest of the game were played at random, and showed that Breaker can keep this expectation from growing. Their argument gives the solution for the colouring theorem and its relatives: the same potential argument, with one player

instead of two, colours the elements one at a time and never lets the expected number of monochromatic sets reach 1.

The argument was recognized as a general derandomization technique, the method of conditional probabilities, systematized by Spencer in 1987 [Spe87] and extended by Raghavan's pessimistic estimators in 1988 [Rag88]. Beck built on the Erdős–Selfridge theorem a theory of positional games, presented in his book of 2008 [Bec08].

*Recognition.* Erdős received the Wolf Prize in 1983/84.

**A.2. The Main Result of this section.** The main result is the Erdős–Selfridge theorem and its two classical consequences.

**Theorem A.1 (Erdős–Selfridge, 1973).** *Let  $\mathcal{F}$  be a family of subsets of a finite set  $V$ .*

- (1) *If  $\sum_{A \in \mathcal{F}} 2^{-|A|} < \frac{1}{2}$ , then in the Maker–Breaker game on  $(V, \mathcal{F})$  in which Maker moves first, Breaker has a winning strategy.*
- (2) *If  $\sum_{A \in \mathcal{F}} 2^{1-|A|} < 1$ , then a colouring of  $V$  in two colours with no monochromatic member of  $\mathcal{F}$  can be computed in time polynomial in  $|V|$  and  $|\mathcal{F}|$ .*
- (3) *In tic-tac-toe on the board  $\{1, \dots, n\}^d$ , the first player cannot force a win when  $\frac{1}{2}((n+2)^d - n^d) < 2^{n-1}$ .*

Part (3) follows from part (1) because the board has  $\frac{1}{2}((n+2)^d - n^d)$  lines of  $n$  cells each: the second player uses Breaker's strategy and occupies a cell of every line. By the Hales–Jewett theorem (1963), on the other hand, the first player wins when  $d$  is large compared with  $n$  [HJ63].

**A.3. The game.** In the *Maker–Breaker game* on  $(V, \mathcal{F})$ , Maker is Player I and Breaker is Player II. Tic-tac-toe is the example to keep in mind:  $V$  is the board and  $\mathcal{F}$  is the set of lines. The players alternately claim unclaimed elements of  $V$ , Player I first, until all elements are claimed. Player I wins if he has claimed all the elements of some  $A \in \mathcal{F}$ , and Player II wins otherwise.

**A.4. How the game solves the problem.** At any stage call  $A \in \mathcal{F}$  *alive* if Player II has claimed none of its elements, let  $u(A)$  be the number of its unclaimed elements, and set

$$\Phi = \sum_{A \text{ alive}} 2^{-u(A)}, \quad w(v) = \sum_{A \text{ alive}, v \in A} 2^{-u(A)}.$$

If the remaining elements were assigned to the two players by fair coin tosses,  $\Phi$  would be the expected number of sets completed by Player I. Initially  $\Phi < \frac{1}{2}$ , and after Player I's first move  $\Phi < 1$ . A claim of  $v$  by Player I increases  $\Phi$  by  $w(v)$ , and a claim by Player II decreases it by  $w(v)$ . Player II always claims an element of maximal weight. Then Player I's next claim increases  $\Phi$  by at most that maximal weight, because weights only decrease when Player II moves, so  $\Phi$  never increases over a pair of moves and stays below 1. If Player I completed a set  $A$ , its term would be  $2^0 = 1$ , which is impossible. For part (2) one player colours the elements one at a time, choosing at each step the colour that does not increase the expected number of monochromatic sets; the expectation starts below 1, never increases, and at the end is an integer, hence 0. This is not just an existence proof: it is an explicit polynomial-time algorithm, short enough to write in a few lines of Python.

The point of view of games adds robustness. A probabilistic argument shows that a random play is good on average; the game shows that a good outcome can be forced one move at

a time, even against an adversary who chooses half of the moves. This is what makes the argument both constructive and efficient.

## APPENDIX B. APPLICATIONS TO FIXED POINT THEORY

**The problem.** The *Brouwer fixed point theorem* says that every continuous map  $f: [0, 1]^n \rightarrow [0, 1]^n$  has a fixed point. The original proofs used algebraic topology. The problem is to find an elementary, combinatorial proof that exposes the finite core of the theorem and, ideally, yields a method for locating approximate fixed points.

✂ **B.1. History of the problem and its solution using combinatorial games.** The theorem was proved in general by Hadamard in 1910 and Brouwer in 1912 [Bro12], by methods of the new algebraic topology. A combinatorial proof was given by Sperner in 1928 [Spe28], through his lemma on labelled triangulations, and Scarf turned this line of thought into an algorithm for computing approximate fixed points in 1967 [Sca67]. Meanwhile the game of Hex was invented by Piet Hein in 1942 and, independently, by John Nash at Princeton in 1948. Nash observed that the game cannot end in a draw and, by a strategy-stealing argument, that the first player has a winning strategy.

The game-theoretic solution was found by David Gale in 1979 [Gal79], who proved that the Brouwer fixed point theorem is an easy consequence of the fact that Hex cannot end in a draw, and extended both statements to  $n$  dimensions.

*Recognition.* Gale's paper, in the *American Mathematical Monthly*, received a Lester R. Ford Award of the Mathematical Association of America in 1980, the year in which Gale shared the John von Neumann Theory Prize. Nash received the Nobel Memorial Prize in Economic Sciences in 1994 and the Abel Prize in 2015.

**B.2. The Main Result of this section.** The main result is the Hex theorem and its consequence, the fixed point theorem in dimension two.

### Theorem B.1 (Gale, 1979).

- (1) The Hex theorem. *If every cell of a  $k \times k$  Hex board is occupied by a stone of Player I or of Player II, then exactly one of the players has a chain of his or her stones joining his or her two sides of the board.*
- (2) *The Hex theorem implies the Brouwer fixed point theorem for  $[0, 1]^2$ .*

Gale proved the corresponding statements for an  $n$ -dimensional version of Hex and  $[0, 1]^n$ .

**B.3. The game.** Hex is played on a rhombus-shaped board of hexagonal cells, and anyone who has played it knows the feeling that somebody *must* win: if you have blocked your opponent completely, you seem to have connected your own sides. It is convenient to describe the board by its centres. The cells are the points  $(i, j)$  with  $1 \leq i, j \leq k$ , and  $(i, j)$  is adjacent to  $(i \pm 1, j)$ ,  $(i, j \pm 1)$ ,  $(i + 1, j + 1)$  and  $(i - 1, j - 1)$ . Player I owns the bottom and top rows,  $j = 1$  and  $j = k$ , and Player II the left and right columns,  $i = 1$  and  $i = k$ . The players alternately place stones on empty cells, Player I first. Player I wins if his stones contain a chain of adjacent cells from bottom to top, and Player II wins if hers contain one from left to right.

**B.4. How the game solves the problem.**

**No draws.** Gale follows the boundary between the two colours. Add an extra row of Player I's stones above and below the board, and an extra column of Player II's on either side. Start at a corner of the enlarged board, between the two colours, and walk along the edges of the hexagons that separate a stone of Player I from one of Player II. At each vertex three cells meet, two of one colour and one of the other, so the walk continues in exactly one way and never returns to a vertex. Being finite, it must leave the board at another corner, and the stones of one colour along one side of the walk form a winning chain. The two players cannot both win, because a top-to-bottom chain and a left-to-right chain would have to cross, and on the hexagonal board crossing chains share a cell.

**Brouwer's theorem.** Suppose  $f = (f_1, f_2): [0, 1]^2 \rightarrow [0, 1]^2$  has no fixed point. By compactness there is  $\varepsilon > 0$  with  $\max_i |f_i(z) - z_i| \geq \varepsilon$  for all  $z$ , and by uniform continuity there is  $\delta < \varepsilon$  with  $|f_i(z) - f_i(z')| < \varepsilon$  whenever  $|z - z'|_\infty \leq \delta$ . Choose  $k$  with  $1/(k-1) \leq \delta$ , and place at each cell  $(i, j)$  the point  $z = ((i-1)/(k-1), (j-1)/(k-1))$ . Give the cell to Player II if  $|f_1(z) - z_1| \geq \varepsilon$ , and to Player I otherwise, in which case  $|f_2(z) - z_2| \geq \varepsilon$ . By the Hex theorem one of the players has a winning chain. Say Player II has a chain from left to right. On the left side  $z_1 = 0 \leq f_1(z)$ , so there  $f_1(z) - z_1 \geq \varepsilon$ ; on the right side  $z_1 = 1 \geq f_1(z)$ , so there  $z_1 - f_1(z) \geq \varepsilon$ . Along the chain the sign changes, so two adjacent cells  $z, z'$  satisfy  $f_1(z) - z_1 \geq \varepsilon$  and  $z'_1 - f_1(z') \geq \varepsilon$ . Adding,  $(f_1(z) - f_1(z')) + (z'_1 - z_1) \geq 2\varepsilon$ , which is impossible because both terms are less than  $\varepsilon$ . A chain of Player I leads to the same contradiction with  $f_2$ .

The point of view of games is that the topological content of the theorem is the statement that a game cannot end in a draw. Read in the contrapositive, the argument shows that for every  $\varepsilon > 0$  some point  $z$  of a sufficiently fine grid satisfies  $|f(z) - z|_\infty < \varepsilon$ , and the fixed point is obtained by compactness. This finite form is the one in which combinatorial proofs of fixed point theorems, from Sperner's lemma to Scarf's algorithm, are used in computation.

## APPENDIX C. APPLICATIONS TO MODEL THEORY

**The problem.** Two structures  $\mathcal{A}$  and  $\mathcal{B}$  in the same signature are *elementarily equivalent*, written  $\mathcal{A} \equiv \mathcal{B}$ , if they satisfy the same first-order sentences. The problem is to find a method, not requiring the inspection of all sentences, for deciding whether two given structures are elementarily equivalent, and, dually, for proving that a given property of structures cannot be expressed by a first-order sentence.

✂ **C.1. History of the problem and its solution using combinatorial games.** Elementary equivalence was introduced by Tarski, whose programme of the 1930s to 1950s of classifying the complete first-order theories of familiar structures required tools for proving that two structures satisfy the same sentences. The back-and-forth method, which goes back to Cantor's theorem of 1895 that any two countable dense linear orders without endpoints are isomorphic, suggested the answer.

The problem was solved by Roland Fraïssé in 1954 [Fra54], who characterized elementary equivalence by systems of partial isomorphisms, and the solution was recast as a game by Andrzej Ehrenfeucht in 1961 [Ehr61], who used it to prove completeness and decidability of theories. In finite model theory, which developed from the 1970s onward in connection with databases and computational complexity, these games became the principal method for proving that properties are not definable in a given logic; Fagin, for instance, used games of this kind in 1975 to show that connectivity of finite graphs is not expressible in monadic existential second-order logic [Fag75].

**C.2. The Main Result of this section.** The main result is the theorem of Fraïssé and Ehrenfeucht. Write  $\mathcal{A} \equiv_n \mathcal{B}$  if  $\mathcal{A}$  and  $\mathcal{B}$  satisfy the same first-order sentences of quantifier rank at most  $n$ .

**Theorem C.1 (Fraïssé, 1954; Ehrenfeucht, 1961).** *Let  $\mathcal{A}$  and  $\mathcal{B}$  be structures in a finite relational signature, and let  $n \geq 0$ . Then  $\mathcal{A} \equiv_n \mathcal{B}$  if and only if Player II has a winning strategy in the game  $\text{EF}_n(\mathcal{A}, \mathcal{B})$  described below. Consequently,  $\mathcal{A} \equiv \mathcal{B}$  if and only if Player II has a winning strategy in  $\text{EF}_n(\mathcal{A}, \mathcal{B})$  for every  $n$ .*

Two standard applications show how the theorem is used. First,  $(\mathbb{Q}, <) \equiv (\mathbb{R}, <)$ . Second, if  $L_m$  denotes the linear order with  $m$  elements, then  $L_m \equiv_n L_{m'}$  whenever  $m, m' \geq 2^n$ ; hence no first-order sentence is true exactly in the finite linear orders of even size, and, by a standard interpretation of graphs in linear orders, connectivity of finite graphs is not first-order expressible.

**C.3. The game.** The game  $\text{EF}_n(\mathcal{A}, \mathcal{B})$  has  $n$  rounds. The names used in the literature say it all: Player I is the *Spoiler*, who wants to expose a difference between the two structures, and Player II is the *Duplicator*, who wants to show that they look alike. In round  $k$ , Player I chooses one of the two structures and an element of it, and Player II answers with an element of the other structure. After  $n$  rounds the players have produced  $a_1, \dots, a_n \in \mathcal{A}$  and  $b_1, \dots, b_n \in \mathcal{B}$ , and Player II wins if the map  $a_k \mapsto b_k$  is a partial isomorphism, that is, if the  $a_k$  and the  $b_k$  satisfy the same atomic formulas.

#### C.4. How the game solves the problem.

**The theorem.** The proof is by induction on  $n$ , and the game is a direct picture of the quantifiers. A sentence of quantifier rank  $n$  is a Boolean combination of sentences  $\exists x \psi(x)$  with  $\psi$  of quantifier rank  $n - 1$ . If  $\mathcal{A} \models \exists x \psi(x)$  but  $\mathcal{B} \not\models \exists x \psi(x)$ , Player I plays a witness  $a \in \mathcal{A}$  of  $\psi$ , and whatever Player II answers in  $\mathcal{B}$  fails  $\psi$ ; by induction Player I wins the remaining  $n - 1$  rounds. Conversely, in a finite relational signature there are only finitely many formulas of each quantifier rank up to equivalence, so the rank- $(n - 1)$  type of an element is expressed by a single formula, and Player II answers each move with an element of the same type. A statement about all first-order sentences is thus replaced by a finite combinatorial task: finding a strategy.

**The applications.** In  $(\mathbb{Q}, <)$  and  $(\mathbb{R}, <)$ , whatever element Player I chooses, Player II can choose an element in the same position relative to the elements already played, because both orders are dense and have no endpoints; this is the back-and-forth argument, and it wins every  $\text{EF}_n$ . For finite linear orders, Player II keeps the following invariant after round  $k$ : any two consecutive chosen elements, including two imaginary elements placed just below the least and just above the greatest element, are at equal distances in the two orders, or at distances that are both at least  $2^{n-k}$ . The intuition: two elements at distance at least  $2^{n-k}$  look infinitely far apart to a formula that has only  $n - k$  quantifiers left to inspect them, since each quantifier can at best halve the distance. When Player I plays inside a long gap, Player II can split the corresponding gap so that the invariant persists, and the invariant guarantees that the final map preserves order. So a property that separates  $L_m$  from  $L_{m'}$  with  $m, m' \geq 2^n$  needs quantifier rank greater than  $n$ ; since the sizes of finite orders are unbounded, parity needs unbounded quantifier rank, and it is not first-order.

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